SWYNEX TECHNOLOGIES

DATA ANALYST INTERNSHIP PROJECT

END-TO-END CUSTOMER SALES ANALYTICS AND REPORTING PROJECT

Data Cleaning | Exploratory Data Analysis | Business Intelligence

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PROJECT NAME : end-to-end-customer-sales-analytics

ORGANIZATION : Swynex Technologies, Pune

SUBMITTED BY : Mr. Shubham Sidhu Lokhande

ROLE : Data Analyst Intern

TOOLS USED : Microsoft Excel | Microsoft Power BI | Python

SUBMISSION DATE : 06/11/2026

Project Link : https://github.com/shubhamlokhande85/end-to-end-customer-sales-analytics

TABLE OF CONTENTS

1. Executive Summary

2. Introduction

3. Problem Statement

4. Project Objectives

5. Dataset Information

6. Tools and Technologies

7. Data Cleaning and Preprocessing

8. Exploratory Data Analysis Using Python

9. Key Performance Indicators

10. EDA Visualizations and Findings

11. Power BI Interactive Dashboard

12. Key Business Insights

13. Business Recommendations

14. Challenges and Learnings

15. Conclusion

16. Project Deliverables and Links

1. EXECUTIVE SUMMARY

This project presents an end-to-end analysis of customer sales data completed as part of a Data Analyst Internship at Swynex Technologies.

The project was carried out in three major stages. First, Microsoft Excel was used to inspect, clean, and standardize the original customer sales dataset by addressing missing values, duplicate records, inconsistent text values, and formatting issues.

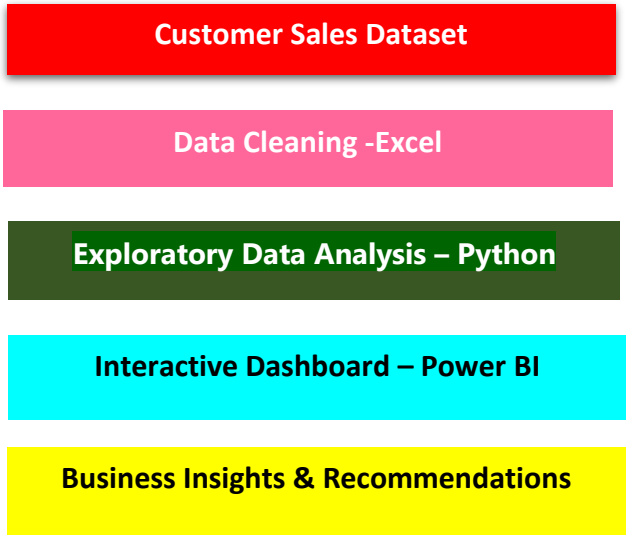
Second, Python was used to perform Exploratory Data Analysis (EDA). Using Pandas, NumPy, Matplotlib, and Seaborn, the dataset was examined to calculate key performance indicators, understand customer purchasing patterns, analyze demographic segments, and identify monthly sales trends.

Finally, Microsoft Power BI was used to develop an interactive dashboard containing KPI cards, charts, and year and month filters. The dashboard provides a visual overview of customer sales performance and purchasing behavior.

The analysis identified useful patterns across gender, age groups, cities, and monthly purchase activity. It also highlighted data-quality concerns that should be considered when interpreting business results.

The final project deliverables include the cleaned dataset, Excel data-cleaning documentation, Python EDA notebook, interactive Power BI dashboard, and this comprehensive project report.

Project Workflow



2.INTRODUCTION

Customer sales analysis plays an important role in understanding purchasing behavior, evaluating business performance, and identifying opportunities for improvement. Customer data contains valuable information about demographics, purchasing patterns, geographical distribution, and sales trends.

This project was completed as part of my Data Analyst Internship at Swynex Technologies. It focuses on analyzing a customer sales dataset using Microsoft Excel, Python, and Microsoft Power BI.

The project follows an end-to-end data analytics approach, beginning with data inspection and cleaning, followed by exploratory data analysis and interactive dashboard development.

Microsoft Excel was used to identify and handle data-quality issues and prepare the dataset for further analysis. Python was used to explore the dataset, calculate statistical measures, identify trends, and create visualizations. Microsoft Power BI was used to develop an interactive dashboard that presents important performance indicators and customer sales insights.

The overall purpose of this project is to transform raw customer sales data into meaningful information that can help businesses understand their customers and make informed decisions.

3.PROBLEM STATEMENT

Businesses generate large amounts of customer sales data containing valuable information about customer demographics, purchasing behavior, and sales performance. However, raw datasets may contain missing values, duplicate records, inconsistent categories, incorrect data formats, and unusual values that can affect the accuracy and reliability of analysis.

Without proper data cleaning and analysis, it becomes difficult to understand customer purchasing patterns, identify important sales trends, compare customer segments, and evaluate business performance.

The objective of this project is to transform a messy customer sales dataset into a structured and useful analytical resource. The project uses Microsoft Excel for data cleaning, Python for exploratory data analysis, and Microsoft Power BI for interactive reporting.

Through this process, the project aims to identify meaningful customer sales patterns, highlight important business insights, and provide recommendations that can support data-driven decision-making.

4.PROJECT OBJECTIVES

The main objective of this project is to analyze customer sales data, identify purchasing patterns, and present meaningful business insights through data analytics and visualization.

The specific objectives are:

1. To inspect and understand the original customer sales dataset.
2. To identify and handle missing values, duplicate records, and inconsistent data.
3. To standardize text values, dates, and numerical formats using Microsoft Excel.
4. To perform exploratory data analysis using Python.
5. To calculate important key performance indicators (KPIs) related to customers, purchases, and feedback.
6. To analyze customer purchasing behavior across gender, age groups, cities, and months.
7. To create meaningful charts and visualizations to identify trends and patterns.
8. To develop an interactive Power BI dashboard with KPI cards, charts, and filters.
9. To identify useful business insights and provide practical recommendations.
10. To document the complete analytics process and present the project through a professional report.

5. DATASET INFORMATION

The dataset used in this project is a customer sales dataset containing information about customer demographics, contact details, registration dates, purchase activity, feedback scores, and geographical information.

The original dataset was provided in CSV format and contained missing values, duplicate records, inconsistent text entries, and certain data-quality issues. It was initially inspected and cleaned using Microsoft Excel before being used for further exploratory analysis and dashboard development.

The same customer sales dataset was used throughout all three stages of the project to maintain continuity between data cleaning, exploratory data analysis, and Power BI dashboard development.

Attribute	Details
Dataset Name	Customer Sales Dataset
Original File Name	messy_customer_sales_dataset.csv
Original Records	10,127
Original Columns	12
Dataset Format	CSV
Dataset Type	Customer and Sales Data
Data Source	Public Dataset

Column Name	Description
Customer ID	Unique identifier assigned to a customer
Names	Customer Names
Gender	Customer gender category
Age	Customer age
City	Customer's city
Signup Date	Customer registration date
Last Purchase Date	Date of the customer's last purchase
Purchase Amount	Recorded customer purchase amount
Feedback Score	Customer feedback rating
Phone Number	Customer contact number
Country	Customer's country

6.TOOLS AND TECHNOLOGIES

This project involved multiple tools and technologies to complete the data analytics workflow. Each tool was selected based on its purpose, from data cleaning and manipulation to statistical analysis, visualization, and interactive reporting.

The combination of these technologies helped transform the original customer sales dataset into a structured analytical resource and present the findings in an understandable format.

Tool / Technology	Purpose	Project Usage
Microsoft Excel	Data cleaning	Identifying missing values, duplicates and inconsistent entries
Python	Data analysis	Performing exploratory data analysis
Pandas	Data manipulation	Loading, inspecting and transforming data
NumPy	Numerical computing	Numerical calculations
Matplotlib	Visualization	Creating analytical charts
Seaborn	Statistical visualization	Creating distribution charts and correlation heatmaps
Jupyter Notebook	Development environment	Writing and executing Python analysis
Microsoft Power BI	Business intelligence	Creating interactive dashboards
GitHub	Version control and hosting	Storing and sharing project files

Technology Workflow

- Microsoft Excel was used for the initial data cleaning and preprocessing.
- Python libraries were used for data manipulation, statistical calculations, exploratory analysis, and visualization.
- Microsoft Power BI was used to create an interactive dashboard with KPI cards, charts, and filters.
- GitHub was selected as the platform for organizing and sharing the final project deliverables.

7.DATA CLEANING AND PREPROCESSING

Data cleaning and preprocessing is an essential stage of the data analytics process. Raw datasets may contain missing values, duplicate records, inconsistent text entries, invalid values, and incorrect formatting that can affect the reliability of analysis.

In this project, Microsoft Excel was used to inspect, clean, and standardize the original customer sales dataset. The objective was to improve data consistency, readability, and usability before performing exploratory data analysis and developing the Power BI dashboard.

The original dataset contained 10,127 records and 12 columns. Various Excel functions and built-in features were used to identify and handle data-quality issues while maintaining a separate copy of the original dataset.

7.1 Data Cleaning Workflow

The following steps were performed to clean and preprocess the customer sales dataset using Microsoft Excel:

1. Imported the original CSV dataset into Microsoft Excel and created a separate working copy to preserve the original data.
2. Reviewed the dataset structure, adjusted column widths, and applied filters to inspect individual columns.
3. Identified duplicate customer IDs or records and removed duplicates using the Data → Remove Duplicates feature.
4. Handled missing Customer ID values using an Unknown_ID placeholder.
5. Standardized Gender values using TRIM(), PROPER(), and Find & Replace to maintain consistent categories.
6. Replaced missing Gender values with Unknown.
7. Cleaned the Age column by separating decimal portions from values such as 51.0 and handled missing ages using an Unknown placeholder.
8. Standardized City values by removing unnecessary spaces and correcting capitalization.
9. Checked Signup Date values for missing entries and applied a consistent date format.
10. Handled missing Last Purchase Date values using an Unknown placeholder and standardized date display.
11. Handled missing Purchase Amount values using a conditional formula.
12. Replaced missing Feedback Score values with 0 and converted formula results into fixed values.
13. Checked Phone Number values using LEN() to identify inconsistent number lengths and applied a length-based standardization rule.
14. Standardized Country values by correcting capitalization and mapping variations such as IND, Ind, India, and INDIA to India.

15. Converted formula-based results into static values using Copy → Paste Special → Values.
16. Created a final working sheet, removed unnecessary columns, and saved the cleaned dataset for further analysis.

7.2 Excel Functions and Features Used

Function / Feature	Purpose	Application
TRIM()	Remove extra spaces	Gender and City
PROPER()	Standardize capitalization	Text categories
IF()	Apply conditional logic	Missing-value treatment
LEN()	Check text length	Phone numbers
Find & Replace	Replace inconsistent entries	Gender and Country
Filters	Inspect records	Identify missing and inconsistent values
Conditional Formatting	Highlight duplicates	Duplicate identification
Remove Duplicates	Remove duplicate records	Dataset cleaning
Text to Columns	Separate decimal portions	Age values
Paste Special → Values	Convert formulas to fixed values	Final cleaning
Number Formatting	Standardize display	Dates and currency

7.3 Missing Value Treatment

Column	Treatment
Customer ID	Unknown_ID
Gender	Unknown
Age	Unknown
City	Unknown
Last Purchase Date	Unknown
Purchase Amount	Conditional formula treatment
Feedback Score	0
Country	Unknown

Missing values were handled using appropriate replacement methods based on the column type and its intended use. Text-based fields were assigned descriptive placeholders such as Unknown or Unknown_ID, while missing feedback scores were replaced with 0. Purchase Amount values were handled using a conditional formula.

Signup Date and Email were checked during the cleaning process and did not contain blank values in the reviewed dataset.

These treatments were applied to improve consistency and prepare the dataset for subsequent analysis. Placeholder values and replaced feedback scores should be considered carefully during statistical calculations to avoid misleading results.

7.4 Before and After Data Cleaning

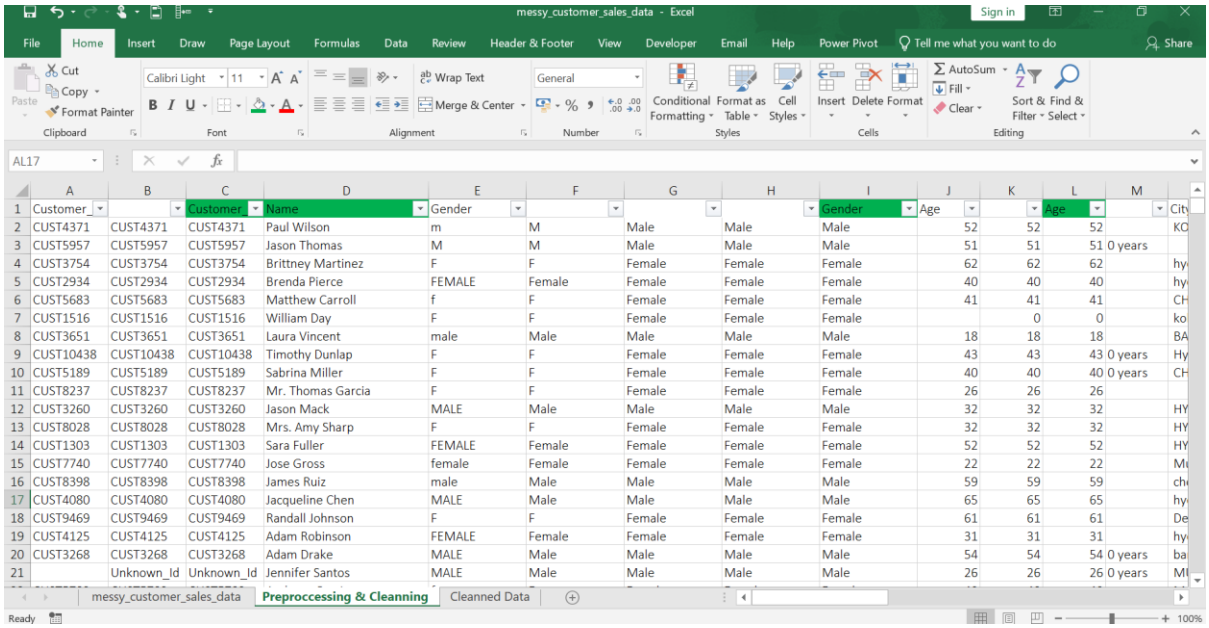
The original dataset contained several data-quality issues, including missing values, duplicate records, inconsistent text formatting, and values requiring standardization.

After the cleaning process, a separate working dataset was prepared with improved consistency and formatting. The cleaned worksheet was saved for use in the next stages of the project, including Python-based exploratory data analysis and Power BI dashboard development.

Original Dataset

	A	B	C	D	E	F	G	H	I	J	K	L
1	Customer	Name	Gender	Age	City	Signup_Da	Last_Purch	Purchase_	Feedback_	Email	Phone_Nu	Country
2	CUST4371	Paul Wilso	m	52	KOLKATA	########	########	26944	1	peckvictor	2.13E+09	India
3	CUST5957	Jason Thor	M	51.0 years		########	########	44152	2	owensanth	1.08E+09	india
4	CUST3754	Brittney M	F	62	hyderabac	########	########	31745	2	tara39@ex	8.98E+09	India
5	CUST2934	Brenda Pie	FEMALE	40	hyderabac	########	########	39674	1	berrynanc	8.23E+09	india
6	CUST5683	Matthew (	f	41	CHENNAI	########	########		8	denise84@	2.67E+09	India
7	CUST1516	William Da	F		kolkata	########	########	27663	10	richard34(	9.46E+09	
8	CUST3651	Laura Vinc	male	18	BANGALO	########	########		3	ymiranda@	6.92E+09	India
9	CUST1043	Timothy D	F	43.0 years	Hyderaba	########	########	18355	9	christophe	9.98E+09	india
10	CUST5189	Sabrina Mi	F	40.0 years	CHENNAI	########	########		10	shepardco	6.23E+09	India
11	CUST8237	Mr. Thoma	F	26		########	########	28206		heidimarti	1.23E+09	InDia
12	CUST3260	Jason Mac	MALE	32	HYDERAB/	########	########	48115	8	campbellrc	8.37E+09	India
13	CUST8028	Mrs. Amy	F	32	HYDERAB/	########	########		8	shannon26	7.02E+09	IND
14	CUST1303	Sara Fuller	FEMALE	52	HYDERAB/	########	########	15663	4	alexanderr	1.17E+09	India
15	CUST7740	Jose Gross	female	22	Mumbai	########	########	41323	2	heather45	5.89E+09	India
16	CUST8398	James Ruiz	male	59	chennai	########	########	39876	4	benjamin1	5.83E+09	India
17	CUST4080	Jacqueline	MALE	65	hyderabac	########	########	18020	7	nicholas29	4.06E+09	InDia
18	CUST9469	Randall Jol	F	61	Delhi	########	########	40633	3	kylecurry@	3.22E+08	India
19	CUST4125	Adam Robi	FEMALE	31	hyderabac	########	########	9091	9	mlang@ex	9.76E+09	India
20	CUST3268	Adam Drak	MALE	54.0 years	bangalore	########	########	3371		simmonsm	3.46E+09	India
21		Jennifer Sa	MALE	26.0 years	MUMBAI	########	########		4	figueroaaa	2.24E+09	India

Cleaning process



Cleaned Dataset

Customer	Name	Gender	Age	City	Signup_D	Last_Purchase_D	Purchase_Amo	Feedback_Sc	Email	Phone_Num	Country
CUST4371	Paul Wilson	Male	52	Kolkata	26-06-2025	17-05-2025	₹ 26,944.00	1	peckvictoria@example.com	2131107701	India
CUST5957	Jason Thomas	Male	51	Unknown	17-02-2021	22-07-2025	₹ 44,152.00	2	owensanthony@example.com	1080761560	India
CUST3754	Brittney Martinez	Female	62	Hyderabad	05-11-2023	08-12-2024	₹ 31,745.00	2	tara39@example.org	8981006345	India
CUST2934	Brenda Pierce	Female	40	Hyderabad	13-03-2022	02-10-2025	₹ 39,674.00	1	berrynancy@example.org	8228064204	India
CUST5683	Matthew Carroll	Female	41	Chennai	05-04-2024	15-12-2024	₹ 0.00	8	denise84@example.org	2665569480	India
CUST1516	William Day	Female	Unknown	Kolkata	02-05-2021	09-06-2025	₹ 27,663.00	10	richard34@example.net	9463089646	Unknown
CUST3651	Laura Vincent	Male	18	Bangalore	16-06-2024	25-04-2025	₹ 0.00	3	ymiranda@example.net	6917110630	India
CUST10438	Timothy Dunlap	Female	43	Hyderabad	16-03-2025	25-04-2025	₹ 18,355.00	9	christopheranderson@example.net	9981404428	India
CUST5189	Sabrina Miller	Female	40	Chennai	07-07-2025	30-10-2024	₹ 0.00	10	shepardcolton@example.net	6233441825	India
CUST8237	Mr. Thomas Garcia	Female	26	Unknown	24-05-2025	15-10-2024	₹ 28,206.00	0	heidimartinez@example.org	1225343854	India
CUST3260	Jason Mack	Male	32	Hyderabad	14-08-2024	20-01-2025	₹ 48,115.00	8	campbellrobert@example.com	8366856220	India
CUST8028	Mrs. Amy Sharp	Female	32	Hyderabad	03-11-2021	09-07-2025	₹ 0.00	8	shannon26@example.org	7022900906	India
CUST1303	Sara Fuller	Female	52	Hyderabad	11-04-2025	29-06-2025	₹ 15,663.00	4	alexandernicholas@example.org	1168031413	India
CUST7740	Jose Gross	Female	22	Mumbai	14-10-2023	24-04-2025	₹ 41,323.00	2	heather45@example.net	5890325488	India
CUST8398	James Ruiz	Male	59	Chennai	07-08-2024	03-09-2025	₹ 39,876.00	4	benjamin18@example.org	5825026489	India
CUST4080	Jacqueline Chen	Male	65	Hyderabad	21-09-2023	11-03-2025	₹ 18,020.00	7	nicholas29@example.com	4062112650	India
CUST9469	Randall Johnson	Female	61	Delhi	25-05-2024	24-02-2025	₹ 40,633.00	3	kylecurry@example.net	0	India
CUST4125	Adam Robinson	Female	31	Hyderabad	16-10-2020	30-09-2025	₹ 9,091.00	9	mlang@example.net	9758358677	India
CUST3268	Adam Drake	Male	54	Bangalore	21-07-2024	24-12-2024	₹ 3,371.00	0	simmonsmorgan@example.com	3458126917	India
Unknown_Id	Jennifer Santos	Male	26	Mumbai	14-11-2022	08-07-2025	₹ 0.00	4	figueroaaaron@example.net	2241159195	India
CUST5782	Anthony Garrison	Female	40	Mumbai	18-10-2024	18-07-2025	₹ 10,907.00	9	kristawade@example.com	0	India
CUST7752	Joshua Hayes	Female	55	Bangalore	07-07-2022	03-03-2025	₹ 45,923.00	0	leejoseph@example.net	8041674560	India
CUST2922	Brittany Young	Male	69	Hyderabad	23-01-2022	26-03-2025	₹ 9,280.00	0	dchavez@example.net	7345041032	India
Unknown_Id	Christopher Harris	Male	Unknown	Hyderabad	27-12-2020	07-07-2025	₹ 42,126.00	10	dawsondeborah@example.org	4001478799	India

7.5 final Outcome

The data cleaning process improved the consistency, readability, and usability of the original customer sales dataset.

The final working worksheet contained the selected columns required for further analysis. The cleaned dataset was then used as the starting point for Python-based exploratory data analysis and interactive Power BI dashboard development.

The cleaning process also highlighted the importance of validating missing-value replacements, customer identifiers, phone number formats, and numerical values before using the dataset for business reporting.

Validation Note

Note: The cleaning process involved placeholder values and conditional replacements. During the subsequent exploratory analysis, additional validation was performed on numerical and date fields. Placeholder customer IDs should not be treated as unique identifiers, and zero feedback scores should be distinguished from genuine customer ratings. Further validation of unusual purchase amounts and invalid age values is recommended before making business decisions.

8.EXPLORATORY DATA ANALYSIS USING PYTHON

8.1 EDA OVERVIEW

Exploratory Data Analysis (EDA) is an important stage of the data analytics process. It involves examining a dataset to understand its structure, identify data-quality issues, calculate statistical measures, and discover meaningful patterns and relationships.

After completing the initial data cleaning process using Microsoft Excel, the customer sales dataset was analyzed using Python. The analysis focused on understanding customer purchasing behavior, demographic distributions, monthly purchase trends, and relationships between numerical variables.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for data manipulation, statistical calculations, and visualization.

The main objective of this stage was to transform the cleaned dataset into meaningful analytical findings that could support business interpretation and the development of the Power BI dashboard.

8.2 Analysis Environment and dataset

Attribute	Details
Project Name	Customer Sales Exploratory Data Analysis
Organization	Swynex Technologies
Dataset	Customer Sales Dataset
Records in Dataset	10,127
Original Columns	12
Analysis Environment	Jupyter Notebook
Programming Language	Python

8.3 Python Libraries and Their Applications

Library	Purpose	Application
Pandas	Data manipulation	Loading, inspecting and transforming datasets
NumPy	Numerical computing	Numerical operations and calculations
Matplotlib	Data visualization	Creating charts and graphs
Seaborn	Statistical visualization	Distribution plots and correlation heatmaps
Jupyter Notebook	Coding environment	Executing Python code and viewing outputs

8.4 Exploratory Data Analysis Workflow

The exploratory data analysis was performed through the following steps:

1. Imported the required Python libraries, including Pandas, NumPy, Matplotlib, and Seaborn.
2. Loaded the customer sales dataset into a Pandas DataFrame.
3. Inspected the first few records using df.head().
4. Checked the dataset dimensions using df.shape.
5. Reviewed column names, data types, and non-null counts using df.info().
6. Generated descriptive statistics using df.describe().
7. Converted the Age column into numeric format and handled invalid and missing values using median imputation.
8. Converted Last_Purchase_Date into the appropriate Pandas datetime format.
9. Standardized inconsistent Gender values using string cleaning methods.
10. Converted four negative Purchase_Amount values into positive values using the absolute-value transformation.
11. Created age groups to compare purchasing behavior across customer segments.
12. Grouped purchase amounts by gender, age group, and month.
13. Calculated important KPIs, including total customers, total purchase amount, average purchase amount, and average feedback score.
14. Created visualizations using Matplotlib and Seaborn.
15. Interpreted the results to identify trends, patterns, anomalies, and potential business insights.

8.5 Data Inspection and Data Transformation

The dataset was inspected using Pandas functions to understand its structure, dimensions, column names, data types, missing values, and statistical distribution.

The Age column was converted into numeric format. Invalid and missing age values were handled using median imputation. The Last_Purchase_Date column was converted into Pandas datetime format to support time-based analysis.

Gender values were standardized by removing unnecessary spaces and converting inconsistent labels into a consistent format.

Four negative purchase amount records were converted into positive values using the absolute-value function. These records were retained for analysis, although their original meaning should be verified before treating the transformed values as confirmed sales.

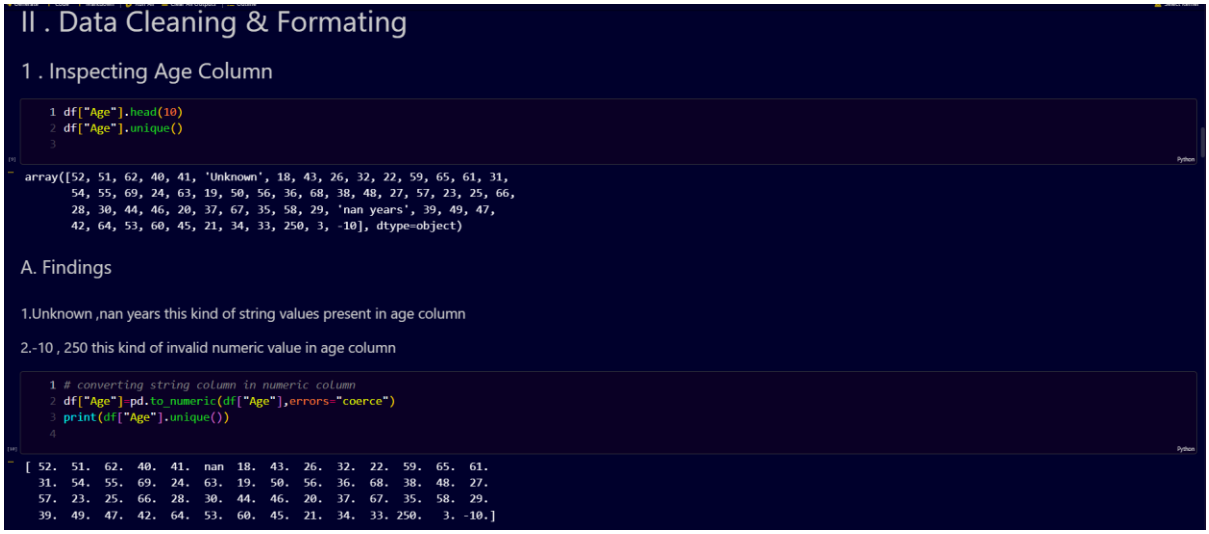
Age groups were created to compare purchase amounts across different customer segments. Monthly purchase amounts were aggregated using the recorded last-purchase dates to understand changes in purchasing activity over time.

These transformations prepared the dataset for statistical analysis and visualization.

Dataset Inspection



Data Transformation



KPI and Statistical Analysis

IV. KPI

```
df.head()
```

	Customer_ID	Name	Gender	Age	City	Signup_Date	Last_Purchase_Date	Purchase_Amount	Feedback_Score	Email	Phone_Number	Country	Age_Group	Months
0	CUST4371	Paul Wilson	Male	52	Kolkata	2025-06-26	2025-05-17	26944	1	peckvictoria@example.com	2131107701	India	45-60	2025-05
1	CUST5957	Jason Thomas	Male	51	Unknown	2021-02-17	2025-07-22	44152	2	owensanthony@example.com	1080761560	India	45-60	2025-07
2	CUST3754	Brittney Martinez	Female	62	Hyderabad	2023-11-05	2024-12-08	31745	2	tara39@example.org	8981006345	India	60+	2024-12
3	CUST2934	Brenda Pierce	Female	40	Hyderabad	2022-03-13	2025-10-02	39674	1	berrynancy@example.org	8228064204	India	30-45	2025-10
4	CUST5683	Matthew Carroll	Female	41	Chennai	2024-04-05	2024-12-15	0	8	denise84@example.org	2665569480	India	30-45	2024-12

```
# 1.Total Customer
print("Total Customer =",df["Customer_ID"].nunique())


#2. Total_Amount
print("Total_Amount =",df["Purchase_Amount"].sum())

#3.Total_Avg_Amount
print("Average_Amount =",df["Purchase_Amount"].mean().round(2))

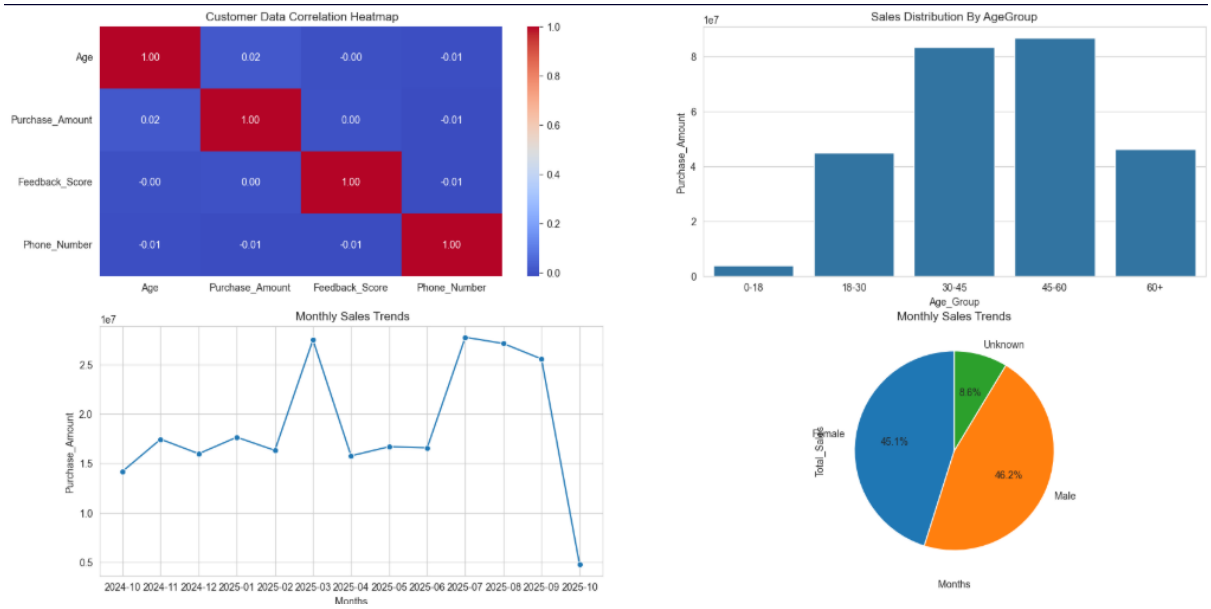
## Average_feedback_Score

print("Average_feedback_Score =",df["Feedback_Score"].mean().round())
```

```
Total Customer = 9001
Total_Amount = 265295613
Average_Amount = 26196.86
Average_feedback_Score = 5.0
```

```
df.to_csv("customer_sales_clean_dataset",index=False)
```

Python Visualizations



8.6 EDA Summary

The exploratory data analysis provided an overview of customer purchasing behavior and sales performance. The analysis identified differences in purchase contributions across gender categories and age groups, highlighted fluctuations in monthly purchase amounts, and examined correlations between selected numerical variables.

The results also revealed data-quality concerns, including invalid age entries, missing purchase dates, negative purchase amounts, and unusually high purchase values. These findings demonstrate the importance of data validation when interpreting analytical results.

The insights generated through Python analysis were used as a foundation for the subsequent Power BI dashboard and business recommendations.

9. KEY PERFORMANCE INDICATORS (KPIs)

Key Performance Indicators (KPIs) are measurable values used to evaluate business performance and understand important aspects of customer activity.

In this project, four major KPIs were calculated using Python to summarize customer numbers, purchase amounts, average spending, and customer feedback.

These indicators provide a high-level overview of the analyzed customer sales dataset and serve as baseline metrics for understanding purchasing behavior and monitoring business performance.

9.1 KPI SUMMARY

KPI	Value
Total Customers	9,001
Total Purchase Amount	26,52,95,613
Average Purchase Amount	26,196.86
Average Feedback Score	5

9.2 KPI Interpretation

The analysis identified 9,001 customers, with a total recorded purchase amount of 265,295,613 and an average purchase amount of 26,196.86. These figures provide an overview of customer activity and overall purchasing performance. The average feedback score was 5.0, indicating a high recorded average. However, missing feedback values replaced with zero and the customer identification method should be validated before drawing final business conclusions. Overall, these KPIs provide a useful summary of the dataset and a starting point for monitoring customer sales performance.

KPI PRESENTATION

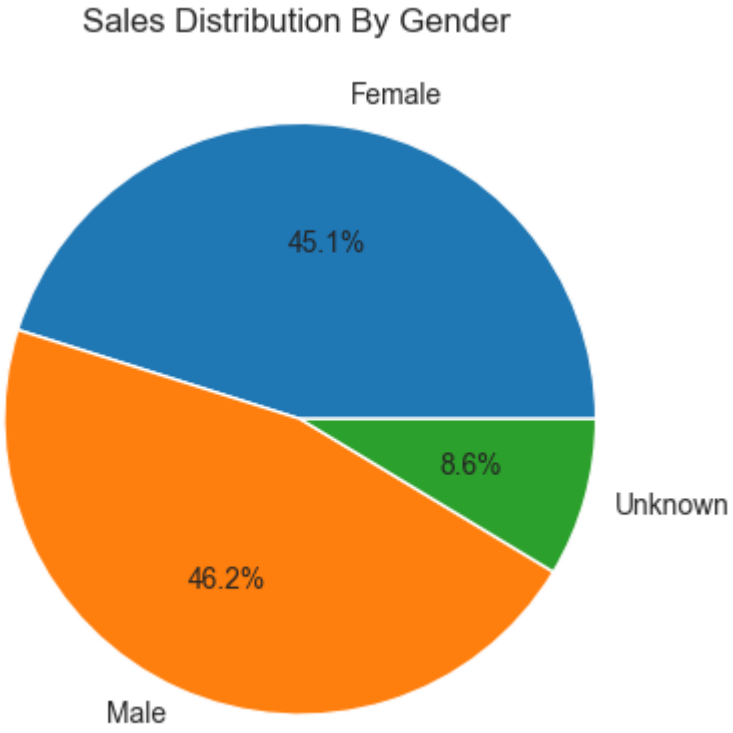
9,001
TOTAL CUSTOMERS
265.3M
TOTAL PURCHASE
26,196.86
AVERAGE PURCHASE
5.0
AVERAGE FEEDBACK

10 . . EDA VISUALIZATIONS AND FINDINGS

Data visualization is an important part of exploratory data analysis because it helps present complex information in an understandable visual format. In this project, Python libraries Matplotlib and Seaborn were used to create charts that explore customer purchasing behavior, demographic contributions, monthly purchase trends, and relationships between numerical variables.

The following visualizations highlight important patterns and provide insights that can support business analysis and decision-making.

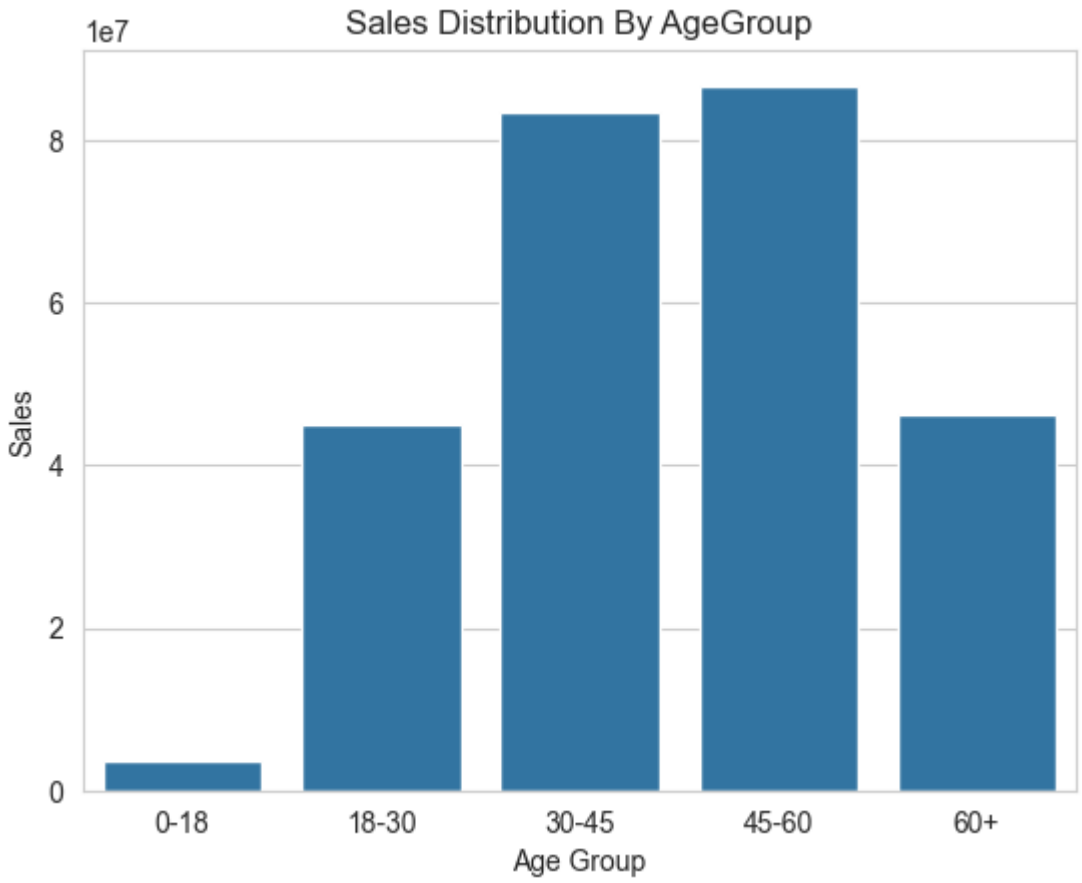
10.1 Purchase Distribution by Gender



The gender-wise purchase analysis shows that male customers contributed approximately 46.2% of the recorded purchase amount, while female customers contributed approximately 45.1%. The Unknown category accounted for approximately 8.6%.

Insight: Male and female customers contributed similar proportions of recorded purchases. Improving missing gender information could help businesses develop more accurate customer segmentation strategies.

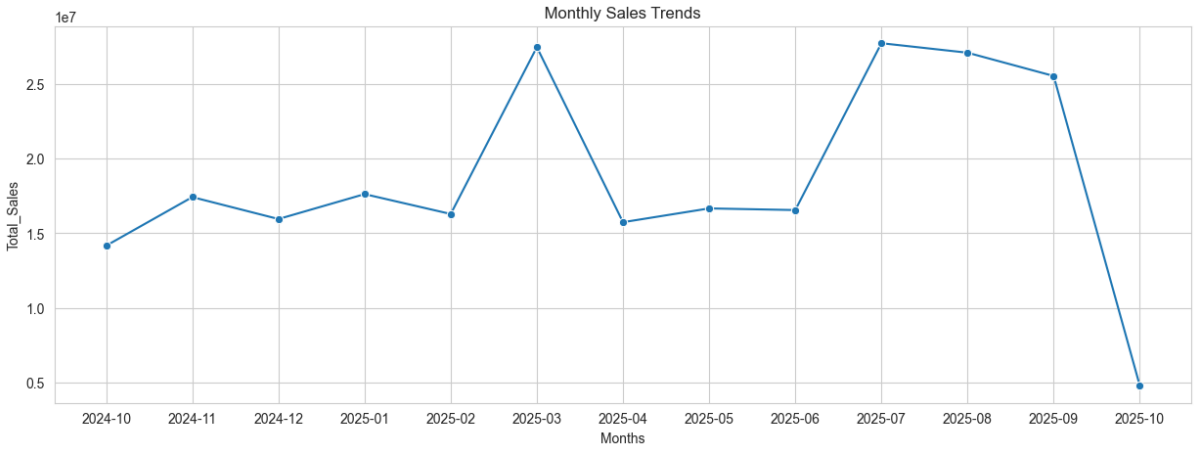
10.2 Sales Distribution By AgeGroup



The age-group analysis shows that customers aged 45–60 contributed the highest total recorded purchase amount among the displayed age groups. The 0–18 age group contributed the lowest total purchase amount.

Insight: The 45–60 age group may represent an important customer segment for further investigation. Comparing average spending and customer counts across age groups would provide a clearer understanding of individual purchasing behavior.

10.3 Monthly Sales Trends



The monthly purchase trend analysis shows fluctuations in recorded purchase amounts between October 2024 and October 2025. Significant increases were observed in March 2025 and July 2025, followed by declines in the subsequent months. October 2025 also showed a substantial decrease compared with September.

Insight: The monthly fluctuations highlight periods that require further investigation. Businesses could examine promotional activities, customer activity, reporting completeness, and unusual transactions to understand these changes.

10.4 Customer Data Correlation Analysis



The correlation heatmap was used to examine linear relationships between selected numerical variables. The correlation between Age and Purchase Amount was approximately 0.02, indicating almost no linear relationship. Feedback Score also showed a correlation close to zero with Purchase Amount.

Insight: Age and feedback score did not show meaningful linear relationships with purchase amount in this analysis. Other customer characteristics and purchasing factors may be more useful for understanding spending behavior.

10.5 Data Quality Issues and Anomalies

During the exploratory analysis, several data-quality issues and unusual values were identified:

- Invalid age entries included -10 and 250.
- The Last_Purchase_Date column contained 1,009 missing values.
- Four records contained negative purchase amounts, which were converted into positive values during preprocessing.
- The maximum recorded purchase amount was 9,999,999 and requires verification.
- Gender categories contained inconsistent formatting before standardization.

Insight: These findings demonstrate the importance of validating unusual values and documenting data-cleaning assumptions before using analytical results for business decisions.

10.6 Overall Key Findings

Analysis Area	Key Finding
Gender	Male and female purchase contributions were similar
Age Group	45–60 had the highest total purchase amount
Monthly Trends	March and July 2025 showed prominent peaks
Correlation	Age and purchase amount had a correlation of approximately 0.02
Feedback	Feedback score showed almost no linear relationship with purchase amount
Data Quality	Missing dates, invalid ages and unusual purchase values were identified

The exploratory analysis revealed useful patterns in customer purchasing behavior, demographic contributions, and monthly purchase activity. It also highlighted data-quality issues that require attention. These findings provide a foundation for the business insights and recommendations presented in the following sections.

11.POWER BI INTERACTIVE DASHBOARD

11.1 Dashboard Overview

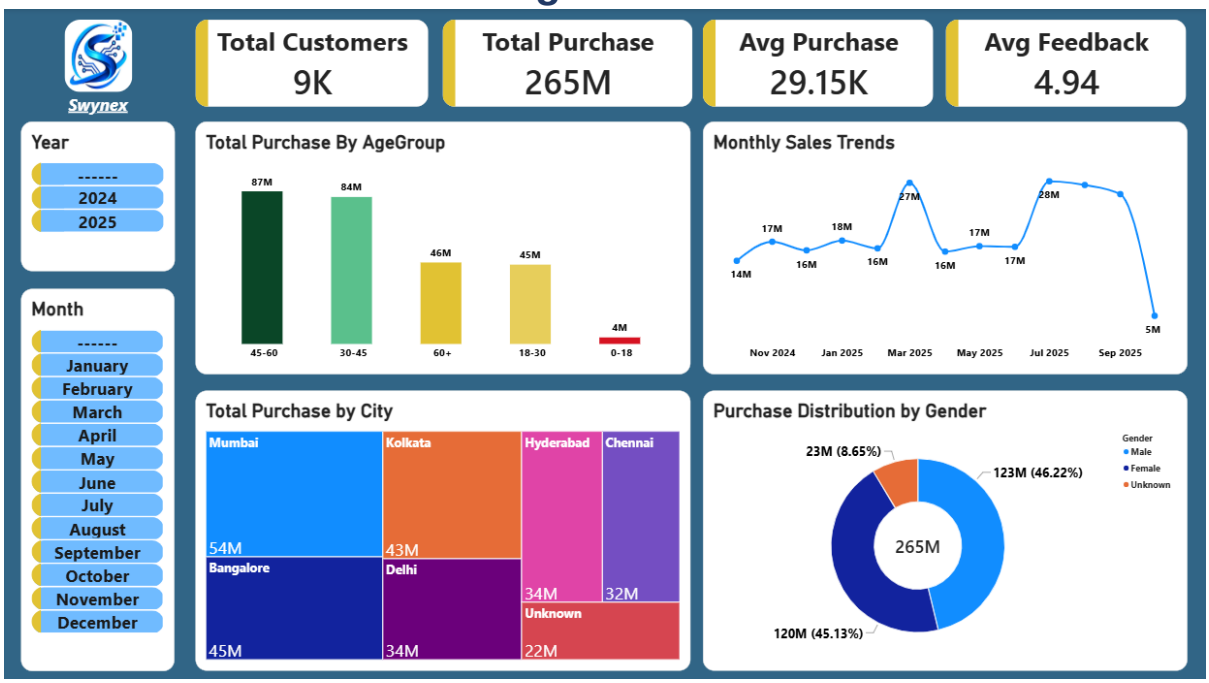
Microsoft Power BI was used to develop an interactive Customer Sales Analysis Dashboard based on the cleaned customer sales dataset.

The dashboard provides a visual overview of customer purchasing behavior, sales performance, demographic distribution, and city-wise purchase contributions.

It combines KPI cards, charts, and interactive slicers to help users explore the data, compare customer segments, and identify important purchasing trends.

The dashboard was designed to present the findings from the Python exploratory data analysis in an easy-to-understand and interactive format.

11.2 Final Dashboard Design



11.3 Key Performance Indicator Cards

The dashboard contains four KPI cards that provide a quick overview of important customer sales metrics:

1. Total Customers – Displays the overall number of customers included in the analysis.
2. Total Purchase – Displays the aggregate recorded purchase amount.
3. Average Purchase – Shows the average purchase amount based on the selected calculation.
4. Average Feedback – Displays the average recorded customer feedback score.

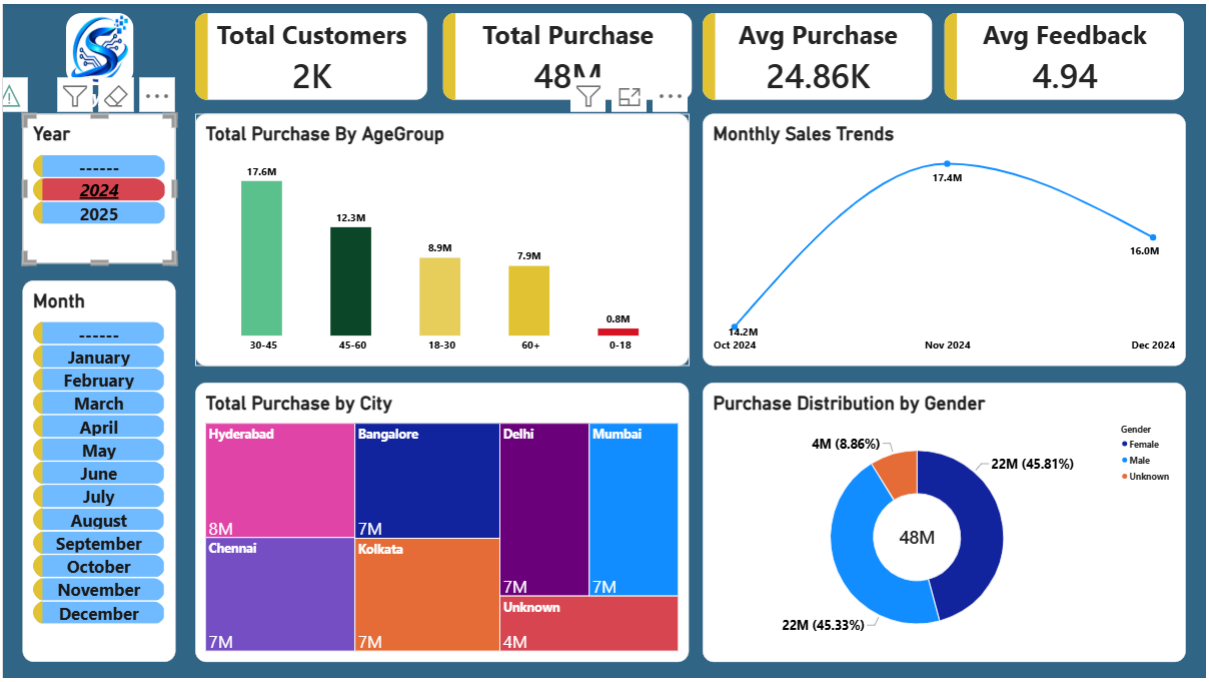
These KPI cards help users monitor important business metrics without manually examining individual records.



11.4 Charts and Visualizations

Visualization	Chart Type	Purpose
Monthly Sales Trends	Line Chart	Identifies changes in monthly purchase amounts
Purchase by Age Group	Column Chart	Compares purchasing contributions across age groups
Purchase by City	Treemap	Shows city-wise purchase contributions
Purchase Distribution by Gender	Donut Chart	Compares purchase contributions by gender

11.5 Interactive Filters



The dashboard includes interactive Year and Month slicers that allow users to filter customer sales information based on selected time periods.

The Year slicer enables users to examine performance across different years, while the Month slicer allows more focused analysis of monthly purchasing activity.

The KPI cards and visualizations respond to the selected filters, making it easier to explore changes in customer sales performance and compare different periods.

11.6 Dashboard Interactivity and Functionality

The Power BI dashboard provides interactive functionality that improves the user experience and supports flexible data exploration.

- Dynamic KPI Cards: Update displayed metrics according to the selected filters.
- Interactive Charts: Allow users to explore different customer sales categories.
- Year and Month Slicers: Filter the dashboard by selected time periods.
- Visual Interactions: Enable users to examine relationships between different dashboard visuals.

These features make the dashboard more useful for exploring customer sales performance and identifying important business patterns.

11.7 Dashboard Outcome

The interactive Power BI dashboard successfully presents important customer sales metrics, demographic comparisons, city-wise purchase contributions, and monthly purchasing trends in a single visual reporting interface.

It transforms analytical results into an accessible format, helping users understand customer behavior, monitor performance indicators, and explore sales information through interactive filters.

The dashboard serves as the business intelligence component of the overall end-to-end customer sales analytics project.

12. KEY BUSINESS INSIGHTS

Business insights are meaningful conclusions derived from analyzing data and identifying patterns that can support better business decisions.

Based on the exploratory data analysis and Power BI dashboard, several useful insights were identified regarding customer purchasing behavior, demographic contributions, monthly purchase trends, geographical performance, and data quality.

These insights help highlight important customer segments, understand changes in purchasing activity, and identify areas that may require further investigation.

12.1 Major Business Insights

Insight 1 – Gender-wise Purchasing Behavior

Male customers contributed approximately 46.2% of the recorded purchase amount, while female customers contributed approximately 45.1%. Both groups showed similar purchasing contributions.

Business Implication: Businesses can consider marketing strategies that appeal to both customer segments rather than focusing exclusively on one gender. Improving missing gender information could also strengthen customer segmentation.

Insight 2 – Age Group Purchasing Patterns

Customers in the 45–60 age group contributed the highest total recorded purchase amount among the analyzed age groups, while the 0–18 group contributed the lowest.

Business Implication: The 45–60 age group could be considered for further customer segmentation and targeted marketing analysis. Customer counts and average spending should also be compared before allocating marketing resources.

Insight 3 – Monthly Purchase Trends

The monthly analysis identified prominent increases in recorded purchase amounts during March 2025 and July 2025, followed by declines in subsequent months. October 2025 also showed a substantial decrease compared with September.

Business Implication: Businesses can investigate these fluctuations by reviewing promotional campaigns, customer activity, transaction records, and reporting completeness. Understanding these patterns may improve future sales planning.

Insight 4 – City-wise Sales Performance

The Power BI treemap visualizes purchase contributions across different cities, helping identify locations with higher and lower recorded purchase amounts.

Business Implication: Businesses can use city-wise comparisons to identify stronger markets and explore opportunities for customer acquisition in locations with lower recorded contributions.

Insight 5 – Data Quality and Customer Insights

The analysis identified missing purchase dates, invalid age entries, inconsistent demographic categories, negative purchase amounts, and an unusually high purchase value of 9,999,999.

Business Implication: Improving data validation and monitoring unusual transactions can increase the reliability of sales reporting and help businesses make better-informed decisions.

12.2 Additional Statistical Findings

Analysis	Finding	Interpretation
Age vs Purchase Amount	Correlation ≈ 0.02	Almost no linear relationship
Feedback vs Purchase Amount	Close to zero	No meaningful linear relationship identified
Gender Distribution	Similar contributions	Both categories contribute comparable purchase amounts
Age Group	45–60 highest total	Important segment for further analysis
Monthly Trends	March and July peaks	Requires investigation

The statistical findings suggest that customer purchase amounts were not strongly associated with age or feedback scores through linear relationships. Therefore, additional factors such as customer frequency, purchase history, product preferences, and location may be useful for deeper customer behavior analysis.

12.3 Business Value of the Analysis

The project demonstrates how customer sales data can be transformed into useful business information through data cleaning, exploratory analysis, and interactive visualization.

The findings provide opportunities to improve customer segmentation, monitor monthly purchasing activity, investigate geographical performance, and strengthen data-quality practices.

The Power BI dashboard makes these findings accessible through interactive filters and visual representations, supporting easier exploration of customer sales information and business performance.

13. BUSINESS RECOMMENDATIONS

Based on the findings from exploratory data analysis and the Power BI dashboard, several recommendations can be suggested to improve customer understanding, monitor sales performance, and strengthen data quality.

These recommendations focus on customer segmentation, monthly sales monitoring, geographical performance, customer feedback, and data validation. They provide possible actions that businesses can consider when planning future sales and marketing activities.

13.1 Recommended Business Actions

1. Improve Customer Segmentation

Since male and female customers contributed similar proportions of recorded purchase amounts, businesses should consider marketing strategies that address the interests of both groups. The 45–60 age group can also be examined further to understand its purchasing preferences and spending behavior.

Recommended Action: Compare customer counts, average purchase amounts, and purchasing frequency across demographic groups before developing targeted campaigns.

2. Monitor Monthly Sales Performance

The analysis identified prominent purchase amount increases during March and July 2025, followed by declines in later months. Businesses should investigate the reasons behind these fluctuations.

Recommended Action: Monitor monthly sales performance, compare promotional periods, and investigate unusual changes to support better sales planning.

3. Explore City-wise Business Opportunities

The Power BI dashboard provides a city-wise comparison of recorded purchase contributions. This information can help identify locations with stronger customer activity and areas that may need additional attention.

Recommended Action: Evaluate customer counts and average purchase amounts by city to identify promising markets and opportunities for customer acquisition.

4. Strengthen Data Quality

Missing values, invalid ages, inconsistent categories, and unusual purchase amounts can affect the accuracy of business analysis.

Recommended Action: Introduce data validation rules, maintain consistent formats, investigate unusual transactions, and regularly review missing and invalid records.

5. Monitor Customer Feedback

The average feedback score was 5.0, but missing feedback values were replaced with zero during data cleaning. Therefore, the reported average should be validated before using it to assess customer satisfaction.

Recommended Action: Track genuine customer feedback separately from missing responses and monitor feedback trends to identify potential service improvements.

6. Track Business KPIs Regularly

The four KPIs developed in this project provide a useful overview of customer numbers, purchase amounts, average spending, and feedback.

Recommended Action: Monitor these metrics regularly through the Power BI dashboard to identify changes in customer activity and overall sales performance.

13.2 Recommendations Summary

Area	Recommended Action	Expected Benefit
Customer Segmentation	Analyze demographic purchasing behavior	Better-targeted marketing
Monthly Sales	Monitor monthly fluctuations	Improved sales planning
City Performance	Compare city-wise customer activity	Identify market opportunities
Data Quality	Introduce validation rules	More reliable analysis
Customer Feedback	Monitor genuine feedback	Better customer experience
KPI Monitoring	Track KPIs regularly	Improved performance monitoring

13.3 Expected Business Impact

Implementing these recommendations may help businesses improve customer segmentation, understand purchasing trends, identify geographical opportunities, and strengthen reporting accuracy.

Regular monitoring of business KPIs and data quality can also support better operational decisions. The interactive Power BI dashboard provides a foundation for tracking these metrics and exploring customer sales performance over time.

These recommendations are based on the patterns observed in the available dataset and should be validated using additional business information before implementation.

14. CHALLENGES AND LEARNINGS

During this end-to-end customer sales analytics project, I gained practical experience in data cleaning, exploratory data analysis, statistical interpretation, and interactive dashboard development.

Throughout the project, I encountered different data-quality and technical challenges that helped me improve my analytical thinking, problem-solving skills, and understanding of the data analytics workflow.

Working with Microsoft Excel, Python, and Power BI provided an opportunity to understand how different tools can be combined to transform raw data into meaningful business insights.

14.1 Project Challenges

1. Handling Missing and Inconsistent Data

The original dataset contained missing values, duplicate records, inconsistent text categories, and formatting issues. Identifying and handling these problems required careful inspection and the use of Excel functions and data-cleaning features.

Learning: I understood the importance of data quality and the need to validate datasets before performing analysis.

2. Handling Incorrect Data Types

Some columns contained values that required conversion into appropriate data types, particularly Age and Last_Purchase_Date. Handling these values correctly was necessary for statistical calculations and time-based analysis.

Learning: I improved my understanding of data types and learned how to use Pandas for data conversion and transformation.

3. Identifying Unusual Values and Anomalies

During Python analysis, I identified invalid ages, negative purchase amounts, missing dates, and unusually high purchase values. These issues required additional investigation and careful interpretation.

Learning: I learned that unusual values should be investigated and documented rather than automatically assumed to be valid or incorrect.

4. Creating Meaningful Visualization

Selecting suitable chart types and presenting analytical findings clearly was another challenge. I used Matplotlib and Seaborn for Python visualizations and Power BI for interactive business reporting.

Learning: I developed a better understanding of how charts can communicate patterns, comparisons, and trends effectively.

5.Developing an Interactive Dashboard

Creating a Power BI dashboard required organizing KPI cards, charts, and interactive filters into a clear reporting layout. Ensuring that the visuals responded appropriately to the Year and Month slicers was an important part of dashboard development.

Learning: I gained practical experience in dashboard design, visual organization, and interactive reporting.

14.2 Technical Skills and Tools Learned

Skill / Technology	Application	Learning Outcome
Microsoft Excel	Data cleaning	Missing-value handling and standardization
Python	Data analysis	Programming-based analytical workflows
Pandas	Data manipulation	Dataset inspection and transformation
NumPy	Numerical analysis	Numerical operations
Matplotlib	Visualization	Creating charts
Seaborn	Statistical visualization	Analyzing distributions and correlations
Power BI	Dashboard development	Interactive reporting and KPI visualization

14.3 Professional Skills Developed

In addition to technical knowledge, this project helped me develop several professional skills:

- Analytical Thinking: Understanding datasets and identifying meaningful patterns.
- Problem-Solving: Investigating data-quality issues and applying appropriate solutions.
- Attention to Detail: Checking missing values, inconsistent categories, and unusual records.
- Data Interpretation: Converting statistical results into understandable business insights.
- Documentation: Preparing structured technical documentation and a professional project report.
- Communication: Presenting analytical findings through visualizations and dashboard reporting.

These skills have strengthened my understanding of the responsibilities and practical workflow of a data analyst.

14.4 Overall Learning Experience

This project provided valuable practical experience in completing an end-to-end data analytics workflow. I learned how to inspect and clean raw datasets, perform exploratory analysis using Python, calculate KPIs, create meaningful visualizations, and develop an interactive dashboard using Power BI.

The project also helped me understand the importance of data validation, accurate interpretation, and clear communication of business findings.

Overall, this experience improved my technical abilities and increased my confidence in applying data analytics concepts to real-world business problems.

15. CONCLUSION

15.1 Overall Project Conclusion

This End-to-End Customer Sales Analytics and Reporting Project provided practical experience in transforming raw customer sales data into meaningful business insights.

The project began with data cleaning and preprocessing using Microsoft Excel, where missing values, duplicate records, inconsistent categories, and formatting issues were identified and addressed. Exploratory Data Analysis was then performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to calculate important KPIs, examine customer purchasing behavior, and identify trends and anomalies.

Microsoft Power BI was used to develop an interactive dashboard featuring KPI cards, monthly purchase trends, age-group comparisons, city-wise purchase contributions, gender distribution, and Year and Month filters.

The analysis revealed similar purchase contributions from male and female customers, higher total purchase amounts among customers aged 45–60, and noticeable fluctuations in monthly purchasing activity. It also highlighted data-quality issues that require further validation before making business decisions.

Overall, this project strengthened my practical knowledge of data cleaning, exploratory analysis, data visualization, business intelligence, and analytical reporting. It demonstrated how multiple data analytics tools can be combined to transform raw data into structured information that supports better business understanding and decision-making.

15.2 End-to-End Project Workflow

Project Stage	Tool Used	Work Completed
Data Cleaning	Microsoft Excel	Identified and handled data-quality issues
Exploratory Analysis	Python	Calculated KPIs and analyzed customer trends
Data Visualization	Matplotlib & Seaborn	Created analytical charts
Dashboard Development	Power BI	Developed an interactive dashboard
Final Reporting	Microsoft Word	Documented findings and recommendations

15.3 Final Project Outcome

The final project successfully combines data cleaning, exploratory data analysis, business intelligence, and documentation into a single analytics workflow.

The major deliverables include:

- A cleaned customer sales dataset.
- Python-based exploratory data analysis and visualizations.
- Calculated business KPIs and analytical findings.
- An interactive Power BI dashboard.
- A final project report documenting the complete process.

These deliverables demonstrate the practical application of data analytics tools and techniques to understand customer sales performance and identify potential business opportunities.

15.4 Personal Learning Outcome

Through this project, I gained hands-on experience working with Excel, Python, and Power BI. I improved my ability to clean datasets, perform statistical analysis, create visualizations, develop interactive dashboards, and communicate findings through professional documentation.

This experience has strengthened my interest in data analytics and provided a foundation for working on more advanced data analysis and business intelligence projects in the future.

16.Project Deliverables

The project materials are uploaded to GitHub to provide a centralized location for accessing the completed work and understanding the analytical process.

16.1 Project files and Deliverables

Deliverable	Description	Tool / Format
Original Dataset	Raw customer sales dataset used for analysis	CSV
Cleaned Dataset	Processed, standardized, and error-free dataset	Excel / CSV
Data Cleaning Documentation	Documentation of data cleaning steps, missing values, duplicate removal, and formatting	Word / PDF
Exploratory Data Analysis (EDA)	Python-based analysis, key performance indicators (KPIs), and visualizations	Jupyter Notebook (.ipynb)
Interactive Dashboard	Customer sales dashboard showing important business insights and performance metrics	Power BI (.pbix)
Final Project Report	Complete project documentation, methodology, analysis, findings, and conclusion	Word / PDF

16.2 Projects Tools And Technologies

- **Microsoft Excel:** Data organization, data cleaning, calculations, and preliminary analysis.
- **Python:** Data processing, statistical analysis, and exploratory data analysis.
- **Pandas and NumPy:** Data manipulation and numerical calculations.
- **Matplotlib and Seaborn:** Data visualization and graphical representation.
- **Power BI:** Interactive dashboards, KPI tracking, and business intelligence reporting.
- **Microsoft Word:** Project documentation and final report preparation.

16.3 LINK

Project Name : end-to-end-customer-sales-analytics

Organazation : Swynex Technologies , Pune

Project Author : Shubham Sidhu Lokhande

Role : Data Analyst Intern

Project Link : https://github.com/shubhamlokhande85/end-to-end-customer-sales-analytics

16.4 Project Credits and Special Thanks !

*I would like to express my sincere gratitude to **Swynex Technologies, Pune**, for providing me with the opportunity, guidance, and support throughout this project.*

I am thankful for the valuable learning experience and practical exposure that helped me improve my technical knowledge and develop my skills in Data Analytics using Excel, Python, and Power BI.

I would also like to acknowledge the support and encouragement received during the completion of this project.

Project Credits:
This project was completed as part of my learning and practical training experience with Swynex Technologies, Pune.

Thank you to Swynex Technologies for being a part of my learning journey and helping me build a strong foundation for my future career in Data Analytics.

Thank You!